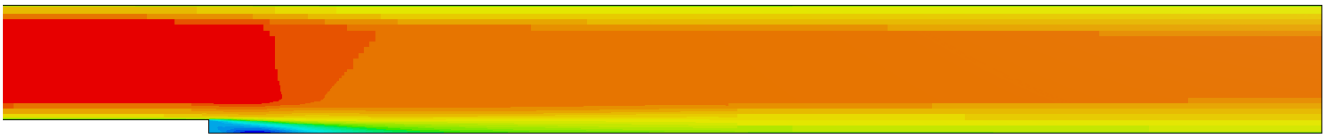
CFD Validation of Backward-Facing Step Flow

Turbulent Flow Simulation Using OpenFOAM, and STAR-CCM+



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March 2026

1 Introduction

The backward-facing step has been widely investigated in both computational and experimental studies by many researchers. Due to its geometrical simplicity and the complex physics involved, it represents one of the most widely used benchmark problems in Computational Fluid Dynamics(CFD). The main phenomena associated with this problem, include flow separation, the formation of the recirculation zone, and flow reattachment downstream of the step. These phenomena can be found in many engineering applications, such as flows over an airfoil and flows in Internal Combustion Engines. One of the key characteristics of the backward-facing step is the reattachment length downstream of the step. Controlling the reattachment length and the adverse pressure gradients that occur near the step is important for optimizing the aerodynamic performance of vehicles, and improving lift characteristics of airfoils.

In this project, the open source CFD software OpenFOAM and the commercial CFD solver STAR-CCM+ are used to simulate the turbulent flow in the backward-facing step, at a Reynolds number 36 000. Particular attention is given to the near-wall mesh resolution in order to adequately capture the boundary layer. k–ω turbulence model has been chosen for this study, as it is proven to be a robust model in flows with adverse pressure gradients and separation.

The velocity profiles, pressure coefficient (C_p), skin friction coefficient (C_f), and Reynolds shear stress ($\overline{u'v'}$) are evaluated and compared with experimental data provided by Seegmiller and Driver [1]. Both solvers are found to be precise in the location where the separated flow reattaches the wall, but STAR-CCM+ gives better prediction of the Reynolds shear stress and more accurate approximation of the experimental profiles.

The objective of the present study is twofold. First, to develop a deeper understanding of the flow physics associated with turbulent separation and reattachment in the backward-facing step. Second, to assess the performance of two CFD solvers, the open-source OpenFOAM and the commercial software STAR-CCM+, by comparing their predictions with experimental measurements.

2 Numerical Setup

2.1 Governing equations

For a 2D incompressible turbulent flow the Reynolds Averaged Navier-Stokes equations(RANS) are solved over the computational domain. These equations include the continuity equation, and the momentum equations in x, and y directions. The differential form of the steady-state incompressible equations that govern the flow in the backward-facing step are given below:

$$\overline{u}\frac{\partial \overline{u}}{\partial x} + \overline{v}\frac{\partial \overline{u}}{\partial y} = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x} + \frac{\partial}{\partial x}\Big[(\nu+\nu_t)\frac{\partial \overline{u}}{\partial x}\Big] + \frac{\partial}{\partial y}\Big[(\nu+\nu_t)\frac{\partial \overline{u}}{\partial y}\Big], \tag{1}$$

$$\overline{u}\frac{\partial \overline{v}}{\partial x} + \overline{v}\frac{\partial \overline{v}}{\partial y} = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial y} + \frac{\partial}{\partial x}\Big[(\nu+\nu_t)\frac{\partial \overline{v}}{\partial x}\Big] + \frac{\partial}{\partial y}\Big[(\nu+\nu_t)\frac{\partial \overline{v}}{\partial y}\Big], \tag{2}$$

$$\nabla \cdot \overline{\mathbf{u}} = \frac{\partial \overline{u}}{\partial x} + \frac{\partial \overline{v}}{\partial y} = 0. \tag{3}$$

The term ν_t appearing in the momentum equations shown represents the turbulent (eddy) viscosity which arises from modeling the Reynolds shear stresses using the Boussinesq hypothesis. According to this hypothesis the Reynolds stress tensor can be related to the mean velocity gradients through an eddy viscosity formulation:

$$-\overline{u_i' u_j'} = \nu_t \left(\frac{\partial \overline{U}_i}{\partial x_j} + \frac{\partial \overline{U}_j}{\partial x_i} \right) - \frac{2}{3} k \, \delta_{ij} \tag{4}$$

where ν_t is the turbulence (eddy) viscosity, k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. The following assumptions are made for the flow:

- Newtonian fluid
- Incompressible flow
- No body forces are considered
- Two dimensional (2D), steady-state simulation

2.2 Domain and Mesh

The geometry that is used for the computations is shown in Figure 1. Flow enters the domain from the left(Inlet part) and flows towards the outlet part and exits in the atmosphere. The height of the inlet is 0.1016 m, and the height of the step is 0.0127 m. The height of the outlet part is 0.1143 m, which gives the ratio of the outlet height to the inlet height(expansion ratio) equal to 1.125.

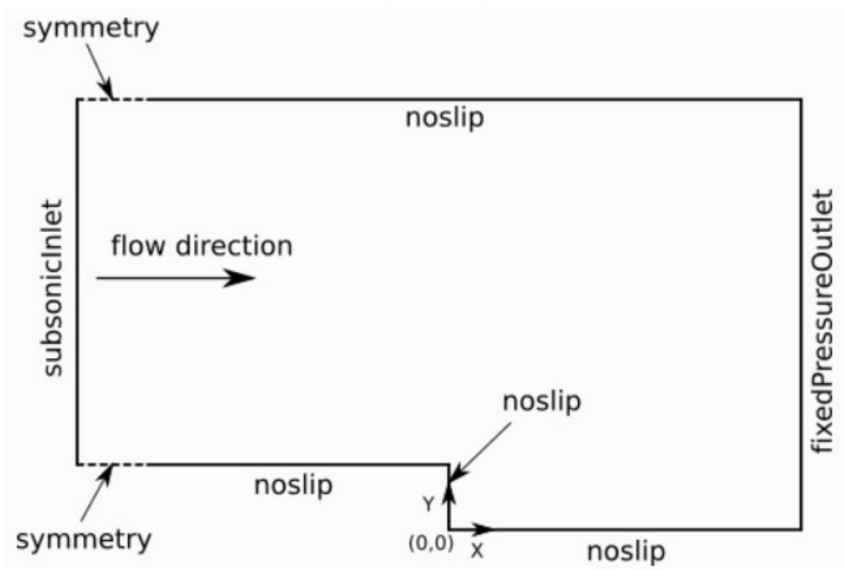

Figure 1: Computational domain of the flow in the backward-facing step. Flow direction from left to right.

In OpenFOAM, the mesh was generated using the BlockMesh utility with structured hexahedral cells. The domain is set as 'empty' in the z-direction to simulate a 2D flow. The mesh is refined near the wall immediately downstream of the step to capture the

recirculation zone and flow reattachment, ensuring that the first cell lies in the viscous sublayer ($y^+ < 1$). The log-law of the wall (Equations 5 and 6) was used to estimate the first cell height, which was adjusted after initial simulations to resolve the steep velocity gradients accurately for the selected turbulence model.

A mesh independence study was conducted using three sequential refinements, resulting in a final mesh of approximately 105,000 cells. Figure 2 shows the final mesh.

$$y^+ = \frac{y\,u_\tau}{\nu} \tag{5}$$

where the friction velocity u_τ is defined as:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \tag{6}$$

with:

- y = distance from the wall to the center of the first cell m
- u_τ = friction velocity m/s
- τ_w = wall shear stress Pa
- ρ = fluid density kg/m^3
- ν = kinematic viscosity m^2/s

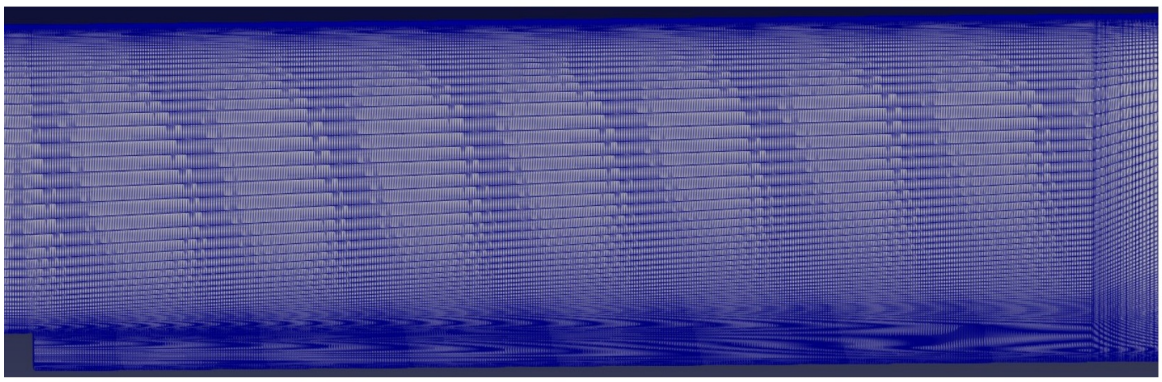

Figure 2: Mesh of the backward-facing step generated in OpenFOAM using the `blockMesh` utility. Structured hexahedral blocks are used, with one cell in the z-direction to simulate a 2D flow. Refinement is applied near the wall to capture the boundary layer and recirculation zone.

The same computational domain and dimensions were recreated in STAR–CCM+ for comparison. A structured mesh was used with refinement near the step and along the lower wall, ensuring $y^+ < 1$ to properly resolve the viscous sublayer and the strong shear layer in the recirculation zone.

2.2.1 Boundary Conditions

The same boundary conditions were applied in both OpenFOAM and STAR–CCM+ simulations. A uniform velocity of 44.2 m/s is imposed at the inlet in the x-direction. Immediately downstream of the inlet, the flow near the top and bottom walls is symmetric; A symmetry boundary condition is applied on these wall sections, while no-slip condition is imposed on the remaining walls. At the outlet, a zero-gradient condition is applied for the velocity (Neumann boundary condition) to allow the flow to exit naturally, and the pressure is fixed at 0 Pa is imposed at the outlet, with zero-gradient condition on the walls to prevent artificial flow penetration.

Turbulence quantities at the inlet were specified using the turbulence intensity $I = 0.061\%$ and the eddy viscosity ratio $\nu_t/\nu = 0.009$. The turbulent kinetic energy k is computed from the inlet velocity U_{in}, and turbulence intensity as:

$$k = \frac{3}{2}\,(U_{\text{in}}\,I)^2$$

where U_{in} is the inlet velocity, and I is the turbulence intensity.

The specific dissipation rate ω is then calculated using the turbulent viscosity ν_t, which can be expressed in terms of the turbulent-to-molecular viscosity ratio ν_t/ν:

$$\omega = \frac{k}{\nu_t}.$$

The chosen values were adopted from the OpenFOAM backward-facing step tutorial, which has been validated against the same experimental data [1]. Several combinations of turbulence intensity and viscosity ratios were tested including (i) Turbulence Intensity + Length Scale, and (ii) Turbulence Intensity + Viscosity Ratio. The final values were selected based on the best agreement between CFD results and the experimental measurements of velocity profiles and skin-friction coefficient C_f.

2.2.2 Turbulence Modeling

The k–ω SST turbulence model is used to model the turbulent flow. This model combines the robust near-wall behavior of the k–ω model with the free-stream independence of the k–ϵ model. It is particularly suitable for flows with adverse pressure gradients and separation, such as the backward-facing step. The eddy viscosity ν_t is computed from the turbulence quantities k and ω, and the Reynolds stresses are modeled using the Boussinesq approximation.

2.2.3 Solver Settings

The simulations were performed using OpenFOAM v11. The steady incompressible Reynolds-averaged Navier–Stokes equations were solved using the `foamRun` application with the `incompressibleFluid` solver module. Pressure–velocity coupling was handled using the SIMPLE algorithm.

The governing equations were discretized using the finite volume method (FVM). In this approach, the computational domain is divided into a set of control volumes, and the governing equations are integrated over each control volume. By applying Gauss' divergence theorem, the volume integrals are converted into surface integrals, resulting in flux balances across the faces of each control volume.

Spatial discretization schemes were selected to ensure numerical accuracy and stability. Gradient terms were computed using the `Gauss linear` scheme, while the convective terms in the momentum equations were discretized using a `linearUpwind` scheme. The convective terms in the turbulence transport equations (`div(phi,k)` and `div(phi,omega)`) were discretized using a bounded `Gauss limitedLinear 1` scheme. Diffusion terms were evaluated using `Gauss linear` interpolation.

Convergence was monitored using the residuals of the governing equations, with tolerances of 10^{-6} for pressure and 10^{-7} for velocity and turbulence variables (k and ω). Under-relaxation factors were applied to improve numerical stability, with a value of 0.9 used for velocity and other transported quantities.

When using STAR-CCM+, the steady-state incompressible Reynolds-averaged Navier-Stokes equations were solved using the segregated flow solver with the SIMPLE pressure-velocity coupling algorithm. Second-order spatial discretization scheme wwas used for thr convective term of the momentum equations. Convergence of the solution was monitored through the residuals of the governing equations as well as the stabilization of the monitored flow quantities.

2.3 Mesh Independence Study

A mesh independence study was conducted in STAR-CCM+ to ensure that the numerical results are independent of the grid resolution. Three different meshes were tested:

- Coarse mesh: 49,678 cells
- Base mesh: 94,535 cells
- Dense mesh: 170,000 cells

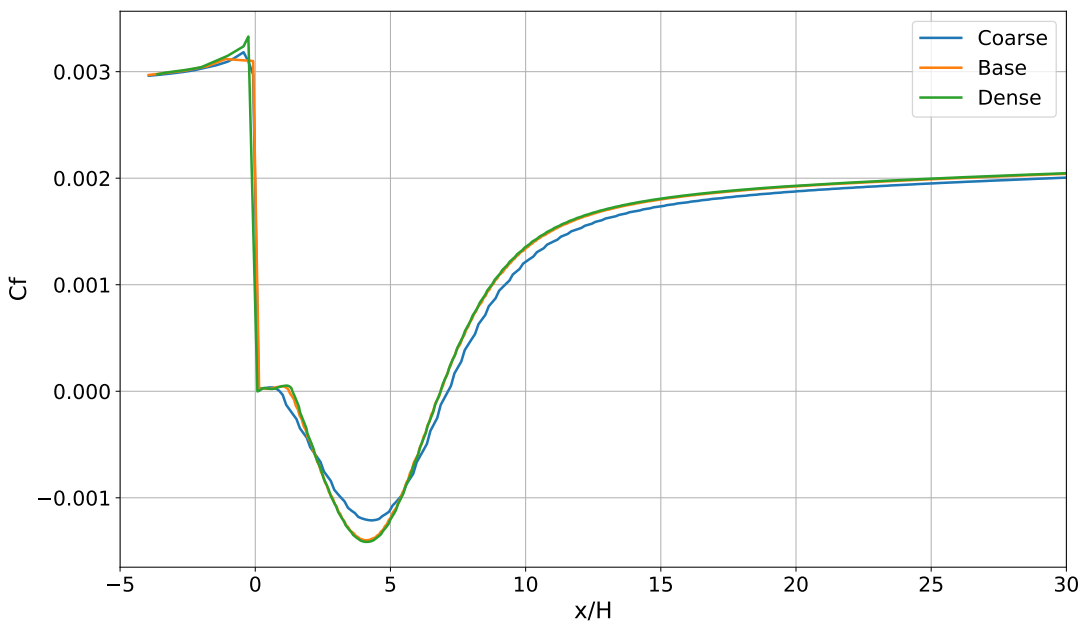

Figure 3: Mesh independence study of the backward-facing step in STAR-CCM+ for three mesh cases: Coarse, Base, and Dense. The skin-friction coefficient C_f along the wall was used to assess mesh convergence and select the Base mesh for the simulations.

Figure 3 shows the skin-friction coefficient C_f along the lower walls for the three different meshes. The base mesh, consisting of 94,535 cells, was selected for the remainder of the study, as the further refinement resulted in small changes in C_f.

2.3.1 Experimental Reference

The experimental study was conducted by Driver and Seegmiller at the NASA Ames Research Center, California. Mean velocity and turbulence measurements were obtained using a Laser Doppler Velocimeter (LDV). Detailed information regarding the measurement techniques can be found in [2].

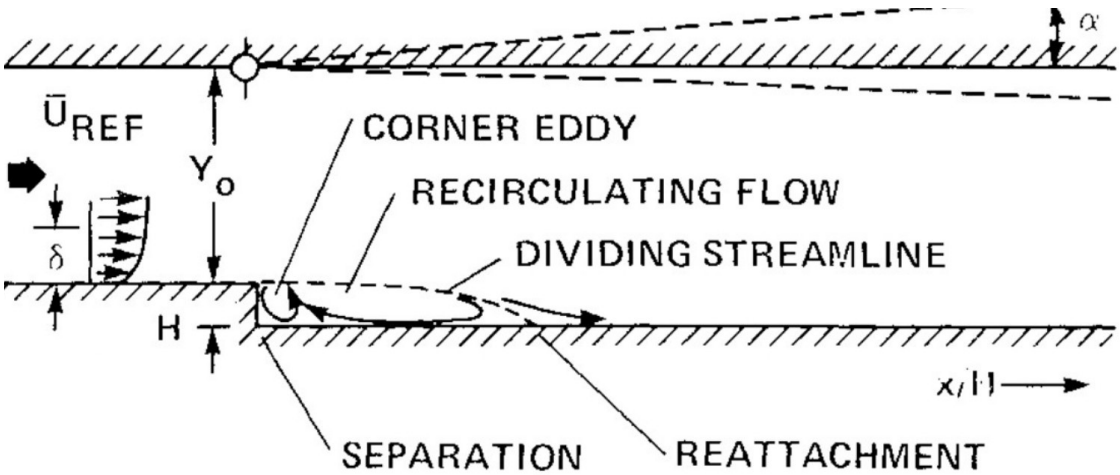

Figure 4: Schematic of the test section used in the experiment of Driver and Seegmiller. Figure taken from [1].

The test configuration consisted of a rectangular duct 1.00 m long, 0.151 m wide, and 0.101 m high, followed by a backward-facing step of 0.0127 m height. The inlet velocity was 44.2 m/s, corresponding to a Mach number of 0.128 and Reynolds number of 36,000 based on the step height. Two cases with different wall divergence angles were investigated in the original study; the present work considers only the case with parallel walls.

Wall-static pressures were measured with 0.2-mm-diameter orifices along the tunnel centerline, typically spaced every half-step height through the recirculation and reattachment zones. The uncertainty in wall-static pressure measurements led to an uncertainty in the wall pressure coefficient of ± 0.009 (95% confidence level).

Time-averaged surface skin-friction measurements of the step-side wall were made using an oil-flow laser interferometer. Skin friction was measured along the step-side wall for both cases of opposite-wall deflection angles. The uncertainties in skin-friction measurements were assessed to be $\pm 8\%$ (95% confidence level), with an increased uncertainty of $+15\%$ in the separated flow region.

The experimental data capture key flow features, including the separation immediately downstream of the step, the recirculation zone, and the reattachment point along the lower wall. These data provide a reliable reference for validation of the present CFD results.

3 Results

In this section, the CFD results for the backward-facing step are presented, including velocity fields, velocity profiles, skin-friction, pressure distribution, and Reynolds shear stress. All post-processing of the CFD data, including visualization and extraction of quantitative values, was performed using ParaView.

3.1 Velocity Field

Figure 5 shows the contours of velocity magnitude over the backward-facing step. A region of low velocity is observed immediately downstream of the step, indicating flow separation. Above the recirculation region, the flow accelerates, reaching the maximum free-stream velocity shown in orange/red in the velocity contours. A key flow characteristic is the reattachment point, located downstream of the recirculation zone, where the wall shear stress becomes zero. In the present study, the reattachment length was found to be approximately $x/H = 6$ in OpenFOAM simulation, while STAR-CCM+ slightly overpredicts it at $x/H = 6.86$. Both predictions are in reasonable agreement with the experimental value of $x/H = 6.26$ reported by Seegmiller and Driver [1].

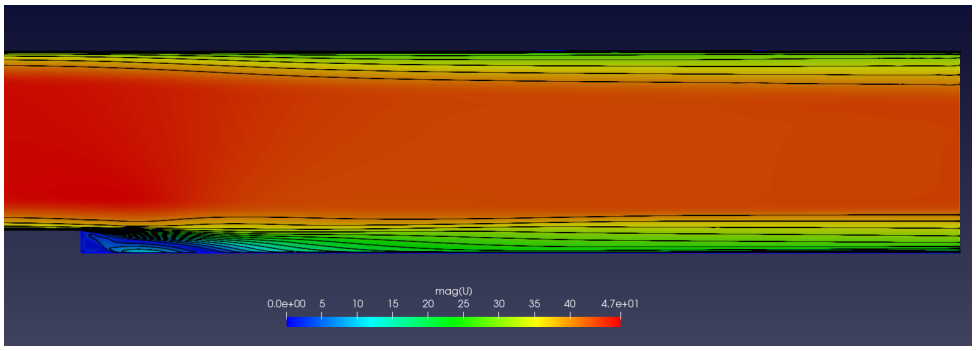

Figure 5: Contours of the velocity magnitude over the backward-facing step from OpenFOAM simulation, showing flow separation and reattachment.

3.2 Velocity Profiles

Figure 6 shows the velocity profiles normal to the wall, computed at four different streamwise positions($x/H = 1, 3, 6, 8$). Both the velocity and wall normal coordinate (y) have been normalized for comparison with the experimental data. At $x/H = 1$, both solvers capture the strong adverse pressure gradient near the step, although small deviations from the experimental data are observed away from the wall. At positions $x/H = 6$ and $x/H = 8$, in the region close to the wall, STAR-CCM+ predicts slightly lower velocities compared to OpenFOAM, indicating a minor momentum deficit. This difference could be related to the near-wall treatment or turbulence modeling in STAR-CCM+. This difference may be attributed to small variations in near-wall mesh resolution. Overall, both solvers reproduce the flow physics and show good agreement with the experimental measurements.

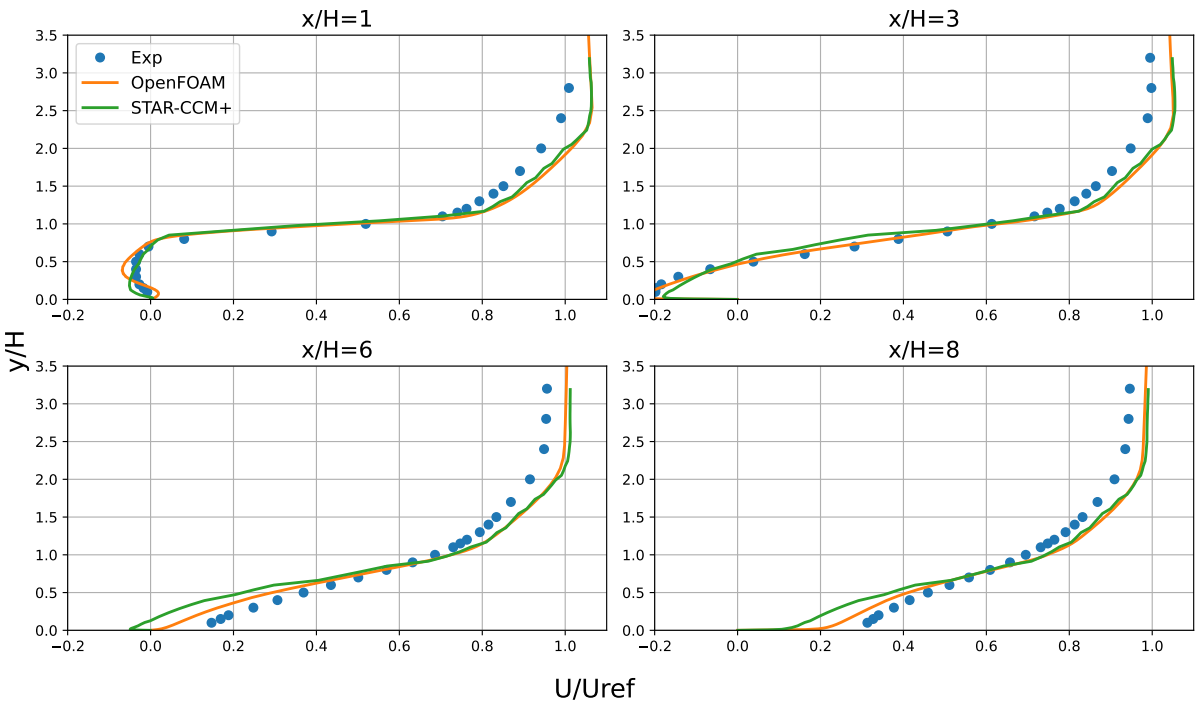

Figure 6: Normalized velocity profiles at streamwise locations $x/H = 1, 3, 6, 8$, obtained from OpenFOAM and STAR-CCM+ compared with the experimental data of Seegmiller and Driver [1].

3.3 Skin Friction Coefficient

Figure 7 shows the skin-friction coefficient C_f along the lower wall, computed from $x/H = -4$ upstream of the step to $x/H = 30$ downstream of the step. This range allows the flow to be examined from the fully developed boundary layer before the step, through the separation region, and further downstream where the flow gradually redevelops.

The skin-friction coefficient is computed from the wall shear stress according to:

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U_{\text{ref}}^2}, \tag{7}$$

where τ_w is the wall shear stress, ρ is the fluid density, and U_{ref} is the reference velocity taken as the inlet velocity.

Both OpenFOAM and STAR-CCM+ capture adequately the flow features: the drop of Cf at the step, the minimum in the separated flow, and the reattachment. Results from STAR-CCM+ agree better upstream of the step , while OpenFOAM predicts the reattachment point at $x/H = 6.26$ closer to the experimental value of $x/H = 6$. A small overshoot around $x/H = 20$–25 is observed in the OpenFOAM results.

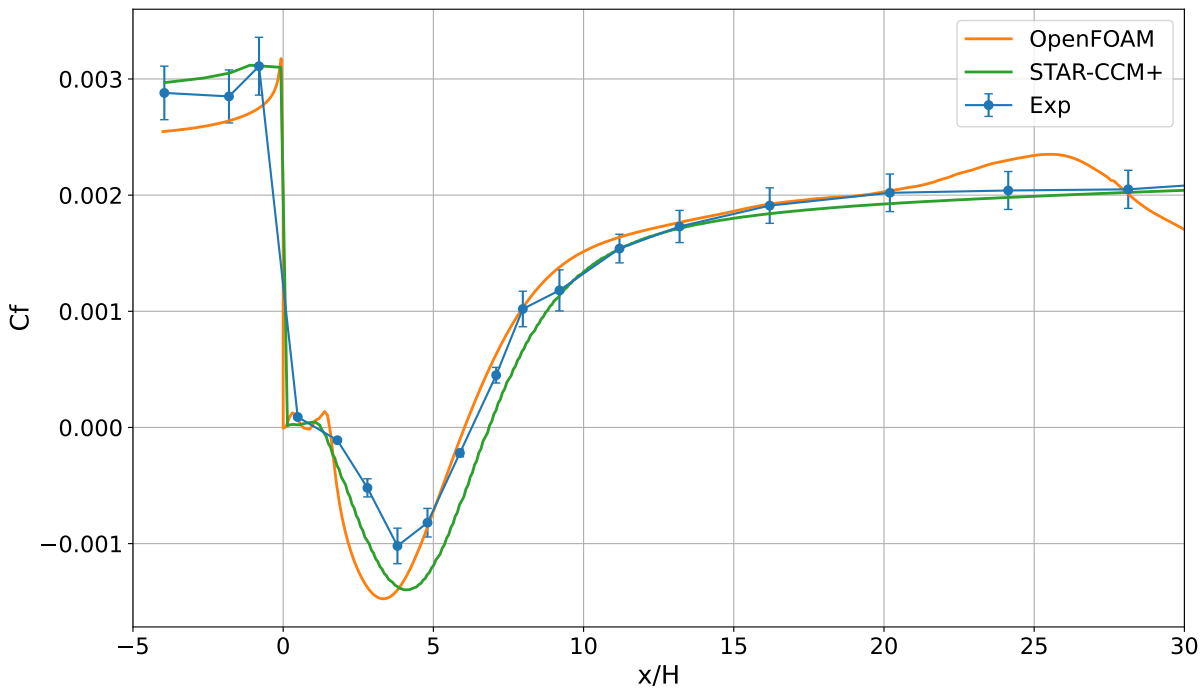

Figure 7: Skin-friction coefficient C_f evaluated on the lower walls upstream and downstream of the step. Results obtained from OpenFOAM, STAR-CCM+ and compared with experimental data.

3.4 Pressure Distribution

Figure 8 shows the pressure coefficient (C_p) along the centerline of the backward-facing step, comparing the CFD results with the experimental data from Driver and Seegmiller [1].

The pressure coefficient is defined as

$$C_p = \frac{p - p_{\text{ref}}}{\frac{1}{2}\rho U_\infty^2}, \tag{8}$$

where p is the static pressure, p_{ref} is the reference pressure (here, the pressure at the outlet, $p_{\text{ref}} = 0$), ρ is the fluid density, and U_∞ is the reference velocity at the inlet.

Upstream of the step ($x/H < 0$), both solvers predict slightly lower C_p values compared to the experiment, with STAR-CCM+ being closer to the experimental data. The differences may be attributed to the assumed inlet boundary conditions and turbulence modeling. Immediately downstream of the step, a sharp drop in C_p is observed due to flow separation. The CFD results capture the overall trend of the pressure recovery in the fully developed region downstream of the reattachment point.

Overall, the agreement between CFD and experimental data is very good beyond the step, with minor deviations observed only in the region upstream of the step.

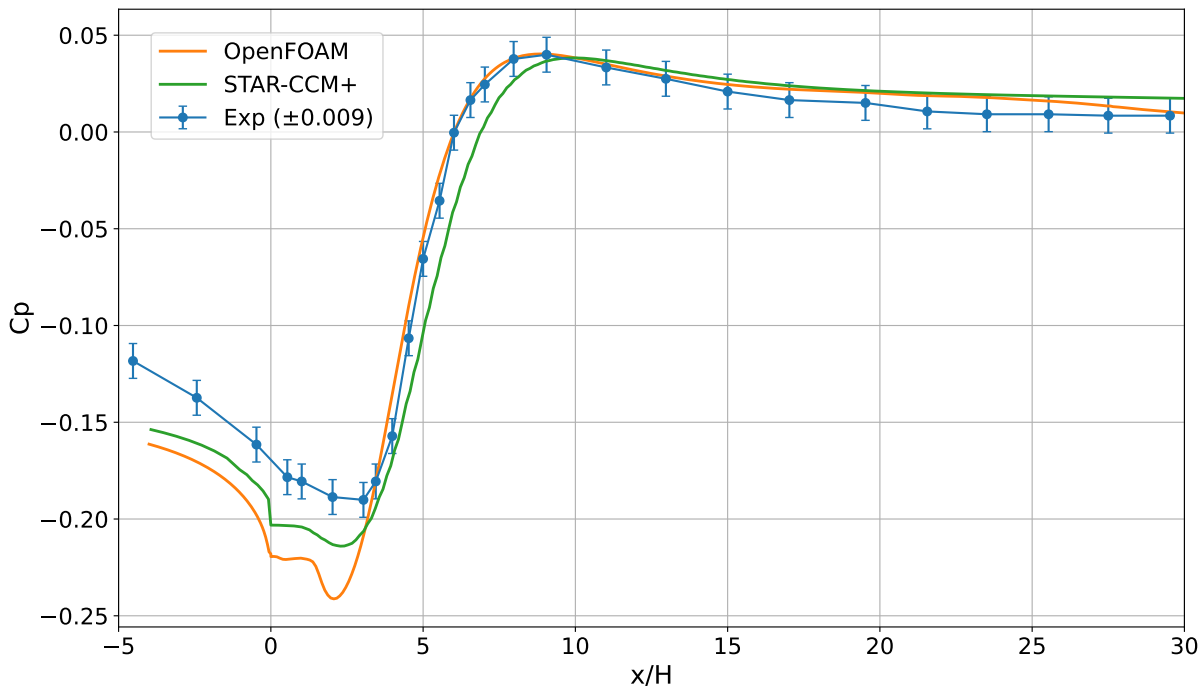

Figure 8: Pressure coefficient C_p evaluated on the lower walls upstream and downstream of the step. Results obtained from OpenFOAM, STAR-CCM+ and compared with experimental data.

3.5 Reynolds Shear Stress

The Reynolds shear stress, $\overline{u'v'}$, is computed using the Boussinesq approximation:

$$-\overline{u_i'u_j'} = \nu_t \left(\frac{\partial \overline{U}_i}{\partial x_j} + \frac{\partial \overline{U}_j}{\partial x_i} \right) - \frac{2}{3} k \, \delta_{ij}, \tag{9}$$

where ν_t is the eddy viscosity, $\overline{U}_i$ and $\overline{U}_j$ are the mean velocity components, k is the turbulent kinetic energy, and δ_{ij} is the Kronecker delta. For the shear stress component $\overline{u'v'}$ computed here, we take $i = 1$ and $j = 2$, so that $\delta_{12} = 0$.

Figure 9 shows the Reynolds shear stress along the backward-facing step at four streamwise locations: $x/H = 1$, 4, 6, and 10.

At $x/H = 1$, OpenFOAM overpredicts the peak of the $\overline{u'v'}$, while STAR-CCM+ underpredicts it. At $x/H = 4$, OpenFOAM is closer to the experimental data than STAR-CCM+, whereas at $x/H = 6$, and $x/H = 10$, STAR-CCM+ provides a better approximation of the experimental measurements.

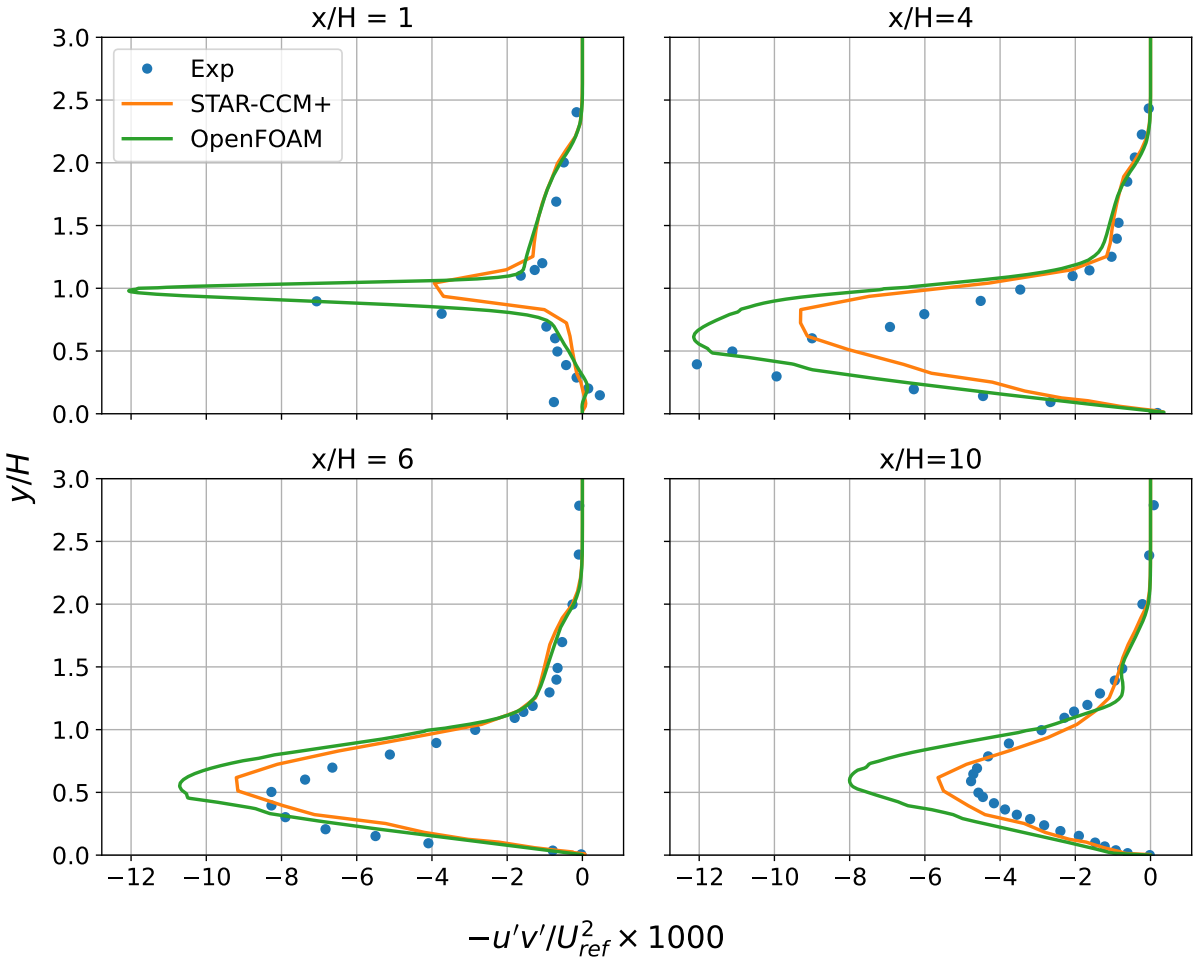

Figure 9: Reynolds Stress $\overline{u'v'}$ at four streamwise locations $x/H = 1$, 4, 6, and 10, obtained from OpenFOAM, and STAR-CCM+ compared by experimental data.

4 Conclusions

The CFD study of the backward-facing step provided detailed insights into the flow behavior, including velocity profiles, skin-friction, pressure distribution, and Reynolds shear stress. The main conclusions are as follows:

- OpenFOAM and STAR-CCM+ adequately captured the separation flow and the recirculation region downstream of the step. The reattachment point was found at $x/H = 6.26$ using OpenFOAM, and slightly higher using STAR-CCM+ at $x/H = 6.86$.

- Velocity profiles along the normal direction at four streamwise locations were measured. Both OpenFOAM and STAR-CCM+ are in good agreement with the velocity profiles obtained from the experiment especially in the area with strong adverse pressure gradients. STAR-CCM+ predicts slightly lower velocities near the wall. Overall, both CFD solvers successfully reproduced the flow acceleration and deceleration within and above the recirculation region.

- The skin-friction coefficient C_f results indicate that STAR-CCM+ approaches better the region upstream of the step, and better approximation in the region after

the reattachment of the flow. Both solvers capture the main features of the flow, including the drop at the step, the minimum friction, and reattachment.

- The pressure coefficient C_p along the centerline shows very good agreement with the experimental data downstream of the step. Upstream of the step and immediately downstream of the step, both solvers slightly underpredict C_p, likely due to differences in inlet boundary conditions and 3D flow effects. Overall, both solvers reproduce the pressure drop and recovery accurately.

- The Reynolds shear stress $\overline{u'v'}$, computed using the Boussinesq approximation, shows varying agreement with the experimental data at different streamwise positions. At $x/H = 1$, OpenFOAM slightly overpredicts the peak while STAR–CCM+ underpredicts it. At $x/H = 4$, OpenFOAM is closer to the experiment, whereas at $x/H = 6$ and $x/H = 10$, STAR–CCM+ provides a better match. These variations in near-wall resolution and the solvers' numerical treatments, highlighting challenges in accurately capturing downstream turbulence structures.

- Overall, the CFD results demonstrate that the model reproduces the main flow features of the backward-facing step and can provide valuable predictions of flow separation, reattachment, and turbulence characteristics, while highlighting areas where turbulence modeling could be improved.

References

[1] David M Driver and H Lee Seegmiller. Features of a reattaching turbulent shear layer in divergent channelflow. *AIAA journal*, 23(2):163–171, 1985.

[2] H Lee Seegmiller, JG Marvin, DR Harrison, and G Kojima. Application of laser velocimetry to an unsteady transonic flow. *AUTOTESTCON 1979*, pages 284–293, 1979.